

# WASP-12b

R-0554

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-12\_171226 · created 2026-07-23T13:01:56+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2458112.84407 +/- 0.00045 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.085 +/- 0.023                   |
| Transit depth (Rp/Rs)^2                           | 0.72 +/- 0.39 [%]                 |
| Orbital Inclination (inc)                         | 86.79 +/- 2.6                     |
| Airmass coefficient 1 (a1)                        | 1.1098 +/- 0.0095                 |
| Airmass coefficient 2 (a2)                        | -0.093 +/- 0.015                  |
| Scatter in the residuals of the lightcurve fit is | 0.85 %                            |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 4.03                              |
| Optimal Annulus                                   | 4.57                              |
| Transit Duration (day)                            | 0.1243 +/- 0.0039                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.085 ± 0.023 vs 0.1178 (deviation 1.41σ) |
| Duration fit vs published | 0.1243 d vs 0.125 d (deviation 0.18σ)     |
| Mid-time O–C              | -1572.9 min                               |
| Depth SNR                 | 3.6                                       |
| Residual scatter          | 0.85 %                                    |

### Light curve

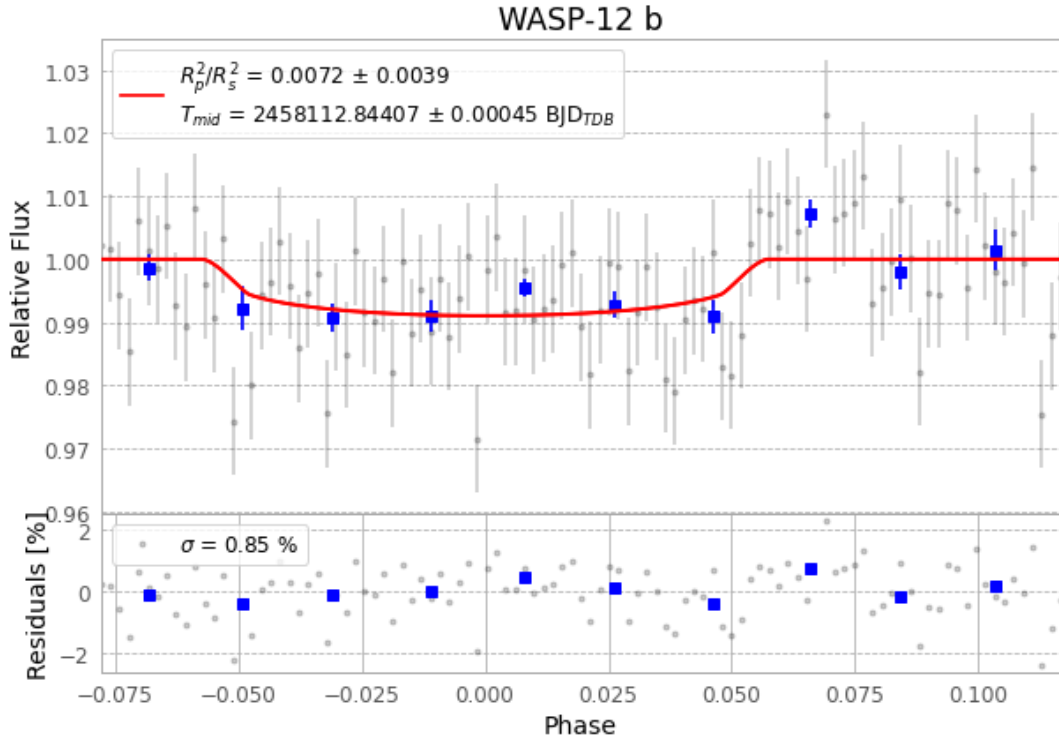


Figure 1 — FinalLightCurve\_WASP-12 b\_2017-12-24.png (SHA-256 aed0e16f8060e901...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 276], 1 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 4128ff7f6e5da603...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit b365494

## Data provenance

106 FITS frames (70 MB), WASP-12171225055412.FITS → WASP-12171225110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-171226.log (0968d226e267...), FinalLightCurve\_WASP-12 b\_2017-12-24.png (aed0e16f8060...), FinalLightCurve\_WASP-12 b\_2017-12-24.csv (0207a5a75f36...)

## Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 13:01Z.